Linear Programming problems

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Outline

Canonical maximization form
- Feasible region and objective function
- Lower bounds
- Graphical solution method

Standard maximization form
- Feasible region and objective function
- Lower bounds
- Graphical solution method

Canonical maximization form

Introduction: Canonical maximization form

- In the chapter «Linear optimization problems» an LP problem is a linear function optimized over a polyhedron: here we write it in a precise form, the **canonical maximization form**
- Its **data** are a vector $\boldsymbol{c}$, a matrix $\boldsymbol{A}$ and a vector $\boldsymbol{b}$: from them we read the feasible region and the objective function
- Every feasible solution gives a **lower bound** on the optimal value $z(\,LP\,)$
- With two variables we can draw everything and solve the problem with the **graphical solution method**: the geometric intuition behind the algorithms of the course

Definition 1: Linear Programming problems in canonical maximization form (1/3)

Given:

1. *an objective-function vector* $\boldsymbol{c} \in \mathbb{R}^n$
2. *a constraint matrix* $\boldsymbol{A} \in \mathbb{R}^{m \times n}$
3. *a right-hand-side (RHS) vector* $\boldsymbol{b} \in \mathbb{R}^m$ *with* m *rows*

$$\underbrace{\begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}}_{\boldsymbol{c}} \in \mathbb{R}^n \quad \underbrace{\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}}_{\boldsymbol{A}} \in \mathbb{R}^{m \times n} \quad \underbrace{\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}}_{\boldsymbol{b}} \in \mathbb{R}^m$$

Definition: Linear Programming problems in canonical maximization form (2/3)

*a **Linear Programming problem** LP in **canonical maximization form** with n non-negative variables*

$$\underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ \vdots \\ x_n \end{pmatrix}}_{\boldsymbol{x}} \in \mathbb{R}^n_{\geq 0}$$

and m less-than-or-equal-to linear constraints is the optimization problem of the next slide.

Definition: Linear Programming problems in canonical maximization form (3/3)

The LP problem in canonical maximization form is the following optimization problem:

$$\max \underbrace{\sum_{j=1}^{n} c_j x_j}_{\boldsymbol{c}'\boldsymbol{x}}$$

$$\textit{subject to} \quad \underbrace{\sum_{j=1}^{n} a_{ij} x_j \leq b_i, \qquad \forall i \in \{1, 2, \ldots, m\},}_{\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}}$$

$$\underbrace{x_j \geq 0, \qquad \forall j \in \{1, 2, \ldots, n\},}_{\boldsymbol{x} \geq 0}$$

where $0 = (0, 0, \ldots, 0)'$ *is a (column) vector of* n *zeros.*

Reading the definition

- The n variables $x_1, x_2, \ldots, x_n$ are **non-negative**
- Every row i of $\boldsymbol{A}$, together with b_i, gives a **less-than-or-equal-to** constraint $\sum_{j=1}^{n} a_{ij}\, x_j \leq b_i$
- The vector $\boldsymbol{c}$ gives the linear function $\sum_{j=1}^{n} c_j\, x_j$ to be **maximized**

Let us now build two LP problems in canonical maximization form from their data $\boldsymbol{c}$, $\boldsymbol{A}$ and $\boldsymbol{b}$.

Example: LP problem in canonical maximization form (1/2)

Consider the following input data of an LP problem with $n = 2$ and $m = 3$:

$$\boldsymbol{c} = \begin{pmatrix} 5 \\ 2 \end{pmatrix} \qquad \boldsymbol{A} = \begin{pmatrix} -1 & 3 \\ 1 & 2 \\ 3 & -2 \end{pmatrix} \qquad \boldsymbol{b} = \begin{pmatrix} 12 \\ 12 \\ 8 \end{pmatrix}$$

The numerical entries of the two column vectors and the matrix are

$$c_1 = 5,\ c_2 = 2,\ a_{11} = -1,\ a_{12} = 3,\ a_{21} = 1,\ a_{22} = 2,$$
$$a_{31} = 3,\ a_{32} = -2,\ b_1 = 12,\ b_2 = 12,\ b_3 = 8.$$

Example: LP problem in canonical maximization form (2/2)

The problem involves $n = 2$ non-negative variables $\boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$.

The associated LP problem in canonical maximization form with $n = 2$ non-negative variables and $m = 3$ less-than-or-equal-to linear constraints reads as follows:

$$
\begin{array}{rrrcrcrcr}
\max & 5 & x_1 & + & 2 & x_2 & & & \\
\text{subject to} & - & x_1 & + & 3 & x_2 & \leq & 12 \\
 & & x_1 & + & 2 & x_2 & \leq & 12 \\
 & 3 & x_1 & - & 2 & x_2 & \leq & 8 \\
 & & x_1, & & & x_2 & \geq & 0
\end{array}
$$

Example: LP problem in canonical maximization form (1/2)

Consider the following input data of an LP problem with $n = 2$ and $m = 2$:

$$\boldsymbol{c} = \begin{pmatrix} -2 \\ -4 \end{pmatrix} \qquad \boldsymbol{A} = \begin{pmatrix} -1 & -1 \\ 1 & -1 \end{pmatrix} \qquad \boldsymbol{b} = \begin{pmatrix} -2 \\ 0 \end{pmatrix}$$

The numerical entries of the two column vectors and the matrix are

$$c_1 = -2,\ c_2 = -4,\ a_{11} = -1,\ a_{12} = -1,$$
$$a_{21} = 1,\ a_{22} = -1,\ b_1 = -2,\ b_2 = 0.$$

Example: LP problem in canonical maximization form (2/2)

The problem involves $n = 2$ non-negative decision variables $\boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$.

The associated LP problem in canonical maximization form with $n = 2$ non-negative variables and $m = 2$ less-than-or-equal-to linear constraints reads as follows:

$$
\begin{array}{rrcrcrr}
\max & -2 & x_1 & - & 4 & x_2 & \\
\text{subject to} & - & x_1 & - & & x_2 & \leq & -2 \\
 & & x_1 & - & & x_2 & \leq & 0 \\
 & & x_1, & & & x_2 & \geq & 0
\end{array}
$$

- Negative data are allowed: here the costs and a right-hand side are negative

Two ingredients

Every LP problem has two ingredients:

the set of points among which we choose,
the **feasible region**,

and the function we maximize,
the **objective function**.

Definition 2: feasible region and objective function (1/2)

- *The **feasible region** of an LP problem in canonical maximization form is the **polyhedron** F defined as follows:*

$$F = \left\{ \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}}_{\boldsymbol{x}} \in \mathbb{R}^n : \underbrace{\sum_{j=1}^{n} a_{ij}\, x_j \leq b_i, \quad \forall i \in \{1, 2, \ldots, m\},}_{\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}} \; \underbrace{x_j \geq 0, \quad \forall j \in \{1, 2, \ldots, n\}}_{\boldsymbol{x} \geq 0} \right\}.$$

The feasible region is defined by the less-than-or-equal-to linear constraints and the non-negativity constraints.

Definition: feasible region and objective function (2/2)

- *The **objective function** of an LP problem in canonical maximization form to be maximized is the (scalar) **linear function** f defined as follows:*

$$f : \mathbb{R}^n \to \mathbb{R}, \qquad f : \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}}_{\boldsymbol{x}} \mapsto \underbrace{\sum_{j=1}^{n} c_j x_j}_{\boldsymbol{c'x}}$$

Let us look at the two ingredients of the first example:
with two variables, everything can be drawn
in the plane (x_1, x_2).

Example: objective function (1/2)

The objective function of the first LP problem in canonical maximization form is the scalar linear function:

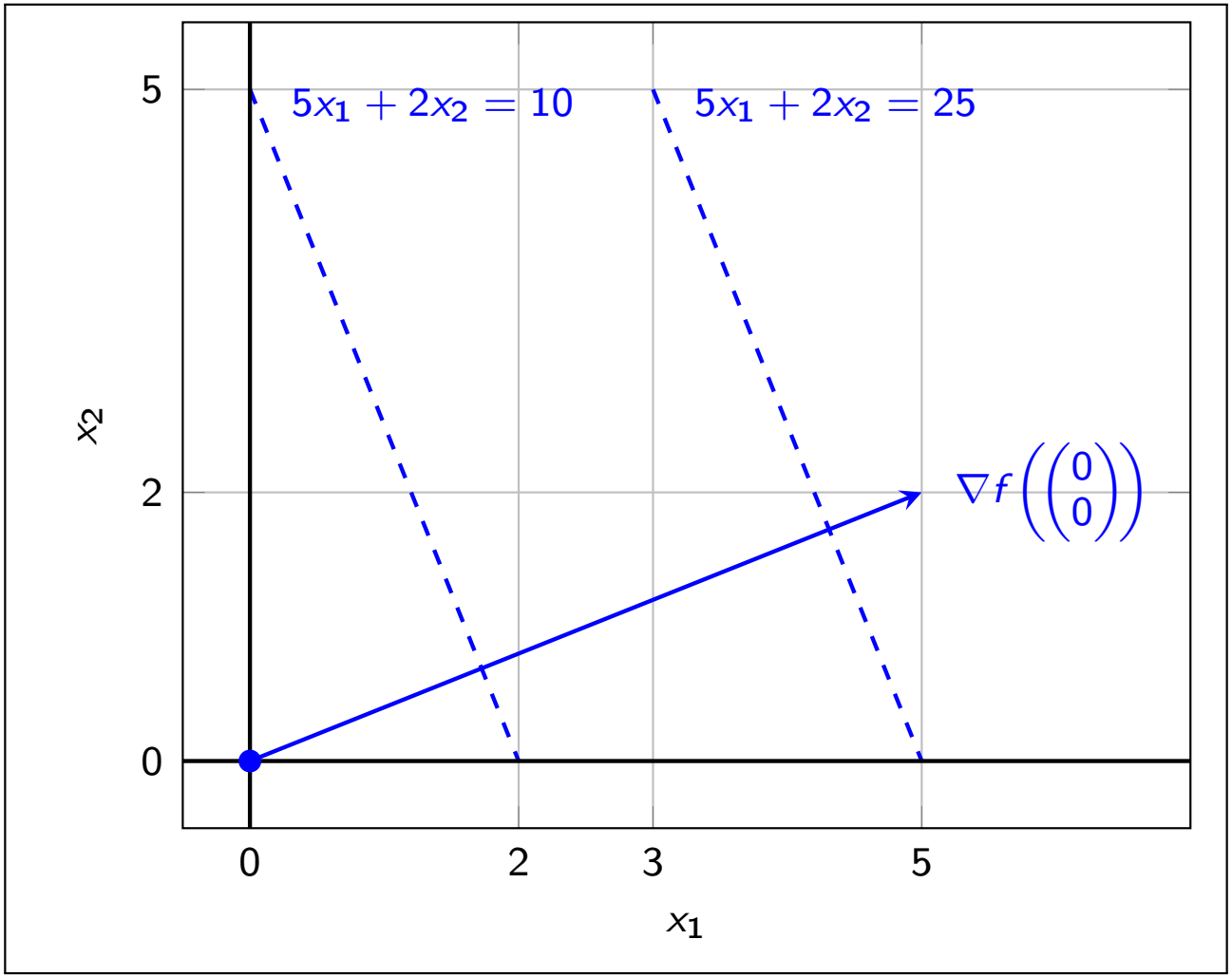
$$f : \mathbb{R}^2 \to \mathbb{R}, \quad f : \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \mapsto 5\,x_1 + 2\,x_2$$

and its gradient is

$$\nabla f \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right) = \begin{pmatrix} 5 \\ 2 \end{pmatrix}.$$

- In the next slide: the gradient drawn from the origin and two level sets of f, the dashed lines $5x_1 + 2x_2 = 10$ and $5x_1 + 2x_2 = 25$, perpendicular to the gradient

Example: objective function (2/2)



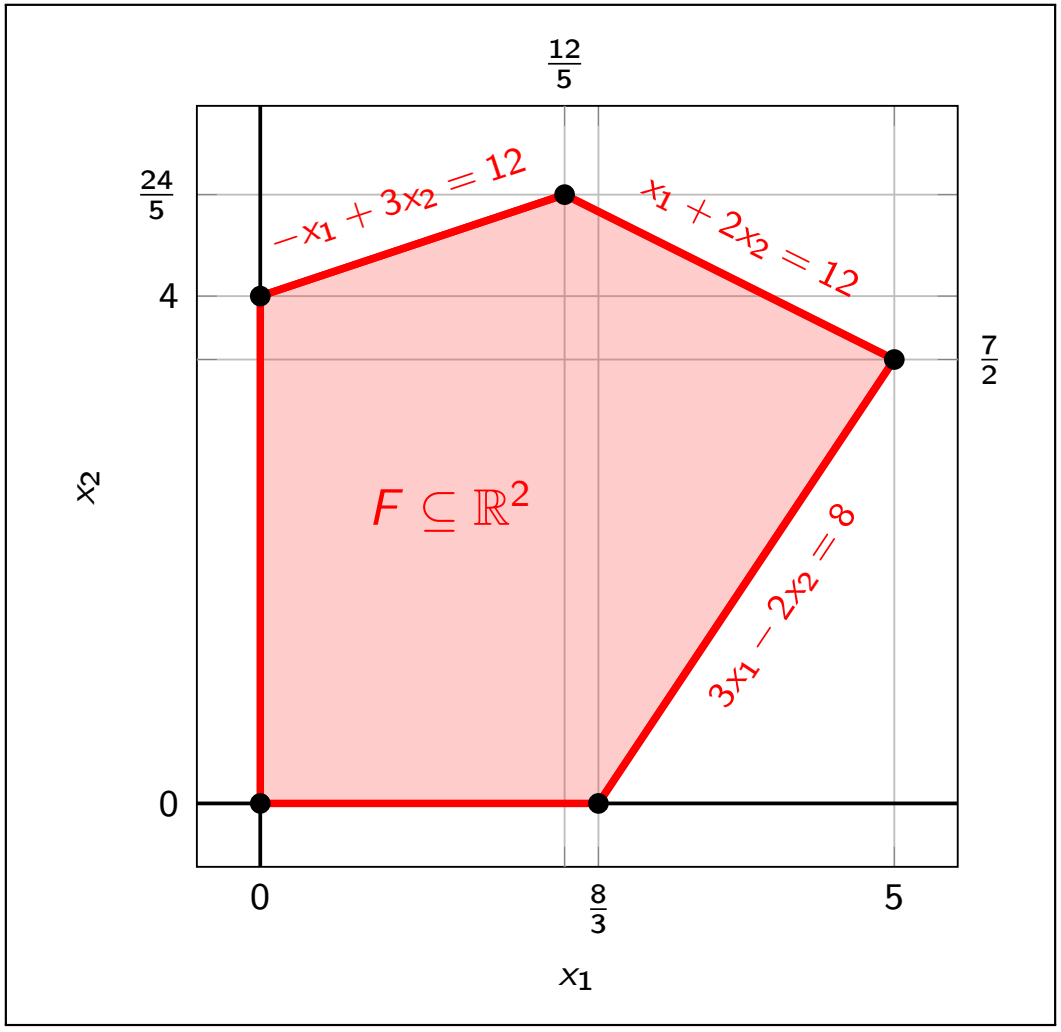
Example: feasible region (1/2)

The feasible region of the first LP problem in canonical maximization form is the polyhedron:

$$F = \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 : \begin{array}{r} -x_1 + 3x_2 \leq 12, \\ x_1 + 2x_2 \leq 12, \\ 3x_1 - 2x_2 \leq 8, \\ x_1,\, x_2 \geq 0 \end{array} \right\}.$$

- In the next slide: F in red, bounded by the three lines $-x_1 + 3x_2 = 12$, $x_1 + 2x_2 = 12$, $3x_1 - 2x_2 = 8$ and by the two axes
- Its vertices (black) are $(0, 0)$, $(0, 4)$, $(\frac{12}{5}, \frac{24}{5})$, $(5, \frac{7}{2})$ and $(\frac{8}{3}, 0)$

Example: feasible region (2/2)



Before solving an LP problem,
every feasible solution already tells us something
about its optimal value $z(\,LP\,)$.

Definition 3: feasible solution and lower bound

- *Given*

$$\underbrace{\begin{pmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \vdots \\ \bar{x}_n \end{pmatrix}}_{\bar{\mathbf{x}}(\,LP\,)} \in F$$

*a **feasible solution** of the LP problem in canonical maximization form (a point satisfying the less-than-or-equal-to constraints and the non-negativity constraints), we can compute a **lower bound** $LB(\bar{\mathbf{x}})$ on the optimal value $z(\,LP\,)$ as follows:*

$$LB(\bar{\mathbf{x}}) = \sum_{j=1}^{n} c_j \, \bar{x}_j$$

Lower bounds and optimal value

For every feasible solution $\bar{\boldsymbol{x}}(\,LP\,) \in F$ of an LP problem in canonical maximization form, we have

$$LB(\bar{\boldsymbol{x}}) \leq z(\,LP\,)$$

- The optimal value $z(\,LP\,)$ is the largest value of the objective function over F: no feasible solution can do better
- A feasible solution with a larger value gives a better (larger) lower bound

Example: feasible solutions and lower bounds (1/2)

Consider the first LP problem in canonical maximization form. A feasible solution is for example

$$\bar{\boldsymbol{x}}(\, LP\,) = \begin{pmatrix} 0 \\ 4 \end{pmatrix} \qquad (\text{where } \bar{x}_1 = 0 \text{ and } \bar{x}_2 = 4)$$

since:

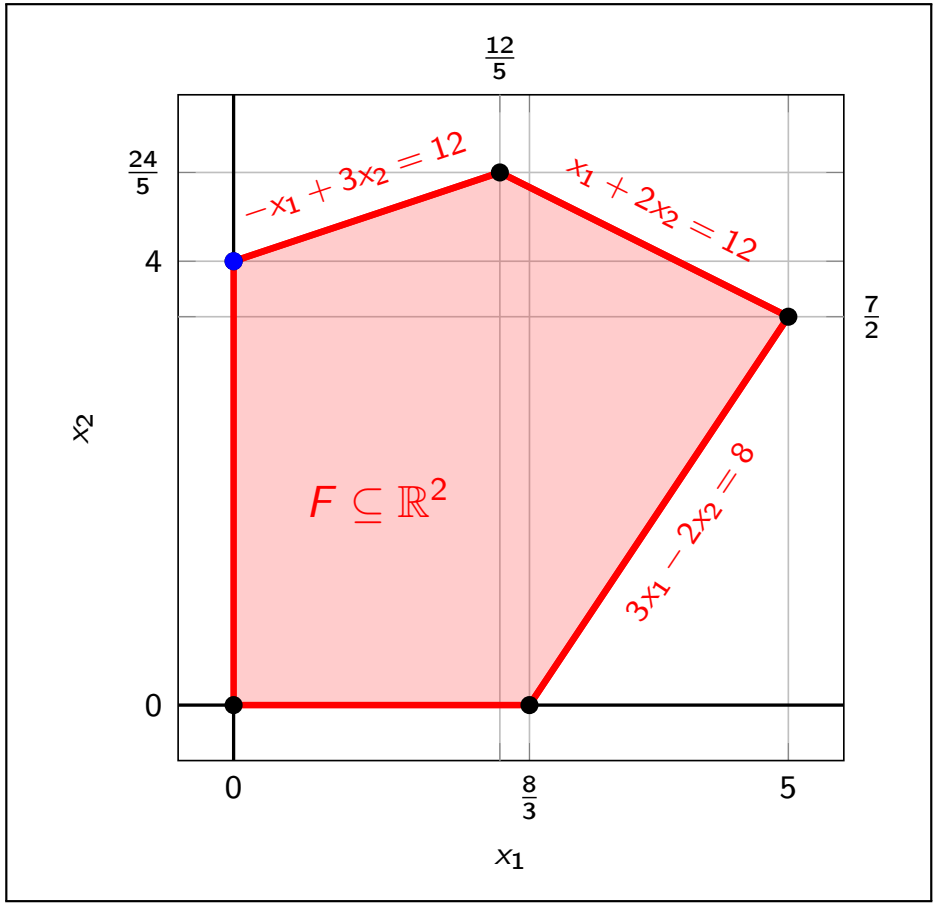
$$\begin{aligned} -0 + 3 \cdot 4 = 12 &\leq 12, \\ 0 + 2 \cdot 4 = 8 &\leq 12, \\ 3 \cdot 0 - 2 \cdot 4 = -8 &\leq 8, \\ 0 \geq 0 \quad &\text{and} \quad 4 \geq 0 \end{aligned}$$

Example: feasible solutions and lower bounds (2/2)

Accordingly, a lower bound $LB(\bar{\boldsymbol{x}})$ on the optimal value $z(\,LP\,)$ of the LP problem is:

$$LB(\bar{\boldsymbol{x}}) = 5 \cdot 0 + 2 \cdot 4 = 8$$

The feasible solution $(0, 4)$ is drawn in blue.



Graphical solution method

An optimal solution of an LP problem in canonical maximization form with two variables can be **graphically** determined using the direction of the objective function gradient and the level sets perpendicular to the gradient.

How the method works

- The level sets $c_1x_1 + c_2x_2 = k$ of the objective function are parallel lines, perpendicular to the gradient $\boldsymbol{c}$
- Moving a level set in the direction of the gradient, the value k grows
- We move it as far as it still touches the feasible region F: the last points of F that it touches are optimal solutions
- We now apply the method to the two examples of this section

Example: graphical solution method (1/3)

For the first LP problem in canonical maximization form:

$$\begin{array}{rrrrrrrr} \max & 5 & x_1 & + & 2 & x_2 & & \\ \text{subject to} & - & x_1 & + & 3 & x_2 & \leq & 12 \\ & & x_1 & + & 2 & x_2 & \leq & 12 \\ & 3 & x_1 & - & 2 & x_2 & \leq & 8 \\ & & x_1, & & & x_2 & \geq & 0 \end{array}$$

an optimal solution is:

$$\underbrace{\begin{pmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{pmatrix}}_{\tilde{\boldsymbol{x}}(\,LP\,)} = \begin{pmatrix} 5 \\ \frac{7}{2} \end{pmatrix}$$

and the optimal value is:

$$z(\,LP\,) = 5\,\tilde{x}_1 + 2\,\tilde{x}_2 = 5 \cdot 5 + 2 \cdot \frac{7}{2} = 32$$

Example: graphical solution method (2/3)

In the figure of the next slide:

- in dashed blue, from left to right, the level sets of the objective function

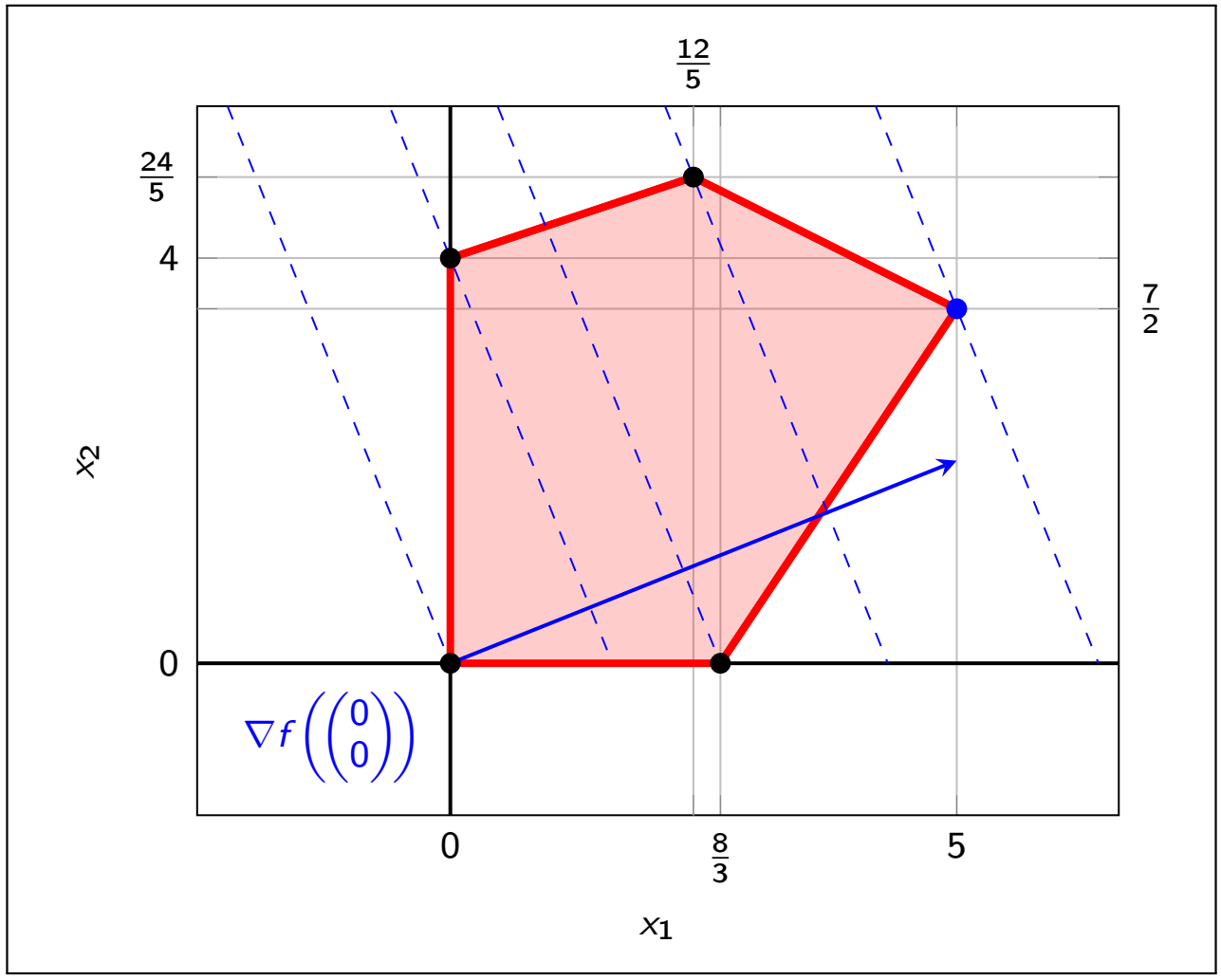
$$5\,x_1 + 2\,x_2 = 0, \qquad 5\,x_1 + 2\,x_2 = 8, \qquad 5\,x_1 + 2\,x_2 = \frac{40}{3},$$

$$5\,x_1 + 2\,x_2 = \frac{108}{5}, \quad 5\,x_1 + 2\,x_2 = 32;$$

  each one passes through a vertex of F: $(0,0)$, $(0,4)$, $(\frac{8}{3},0)$, $(\frac{12}{5},\frac{24}{5})$ and $(5,\frac{7}{2})$

- the gradient $\nabla f = (5,2)'$ is drawn from the origin
- the optimal vertex $(5,\frac{7}{2})$ is depicted in blue: it is the last point of F touched by the level sets moving in the direction of the gradient

Example: graphical solution method (3/3)



Example: graphical solution method (1/3)

For the second LP problem in canonical maximization form:

$$\begin{array}{rrcrcrcr} \max & -2 & x_1 & - & 4 & x_2 & & \\ \text{subject to} & - & x_1 & - & & x_2 & \leq & -2 \\ & & x_1 & - & & x_2 & \leq & 0 \\ & & x_1, & & & x_2 & \geq & 0 \end{array}$$

an optimal solution is:

$$\underbrace{\begin{pmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{pmatrix}}_{\tilde{\boldsymbol{x}}(\,LP\,)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

and the optimal value is:

$$z(\,LP\,) = -2\,\tilde{x}_1 - 4\,\tilde{x}_2 = -2 \cdot 1 - 4 \cdot 1 = -6$$

Example: graphical solution method (2/3)

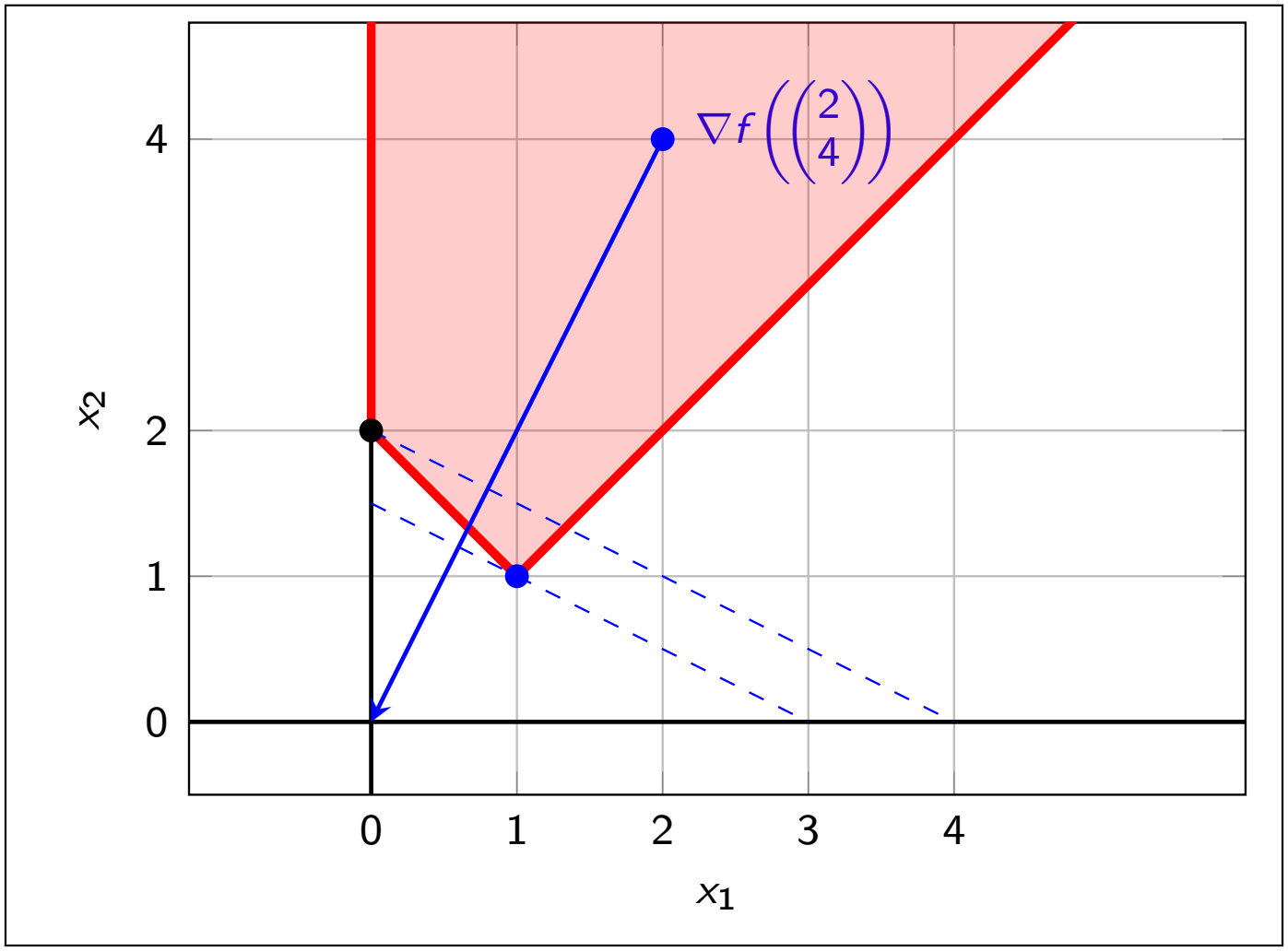
In the figure of the next slide:

- the feasible region F (red) is **unbounded**: it is bounded by the lines $-x_1 - x_2 = -2$ and $x_1 - x_2 = 0$ and by the axis $x_1 = 0$, and it has the two vertices $(0, 2)$ and $(1, 1)$
- in dashed blue, from bottom to top, the level sets of the objective function

$$-2\,x_1 - 4\,x_2 = -6 \qquad \text{and} \qquad -2\,x_1 - 4\,x_2 = -8$$

- the gradient $\nabla f = (-2, -4)'$ is drawn from the point $(2, 4)$: it points towards the origin
- the optimal vertex $(1, 1)$ is depicted in blue

Example: graphical solution method (3/3)



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Summary (1/2)

Canonical maximization form:

- Data: $\boldsymbol{c} \in \mathbb{R}^n$, $\boldsymbol{A} \in \mathbb{R}^{m \times n}$, $\boldsymbol{b} \in \mathbb{R}^m$
- max $\boldsymbol{c}'\boldsymbol{x}$ subject to $\boldsymbol{A}\,\boldsymbol{x} \leq \boldsymbol{b}$, $\boldsymbol{x} \geq 0$: n non-negative variables and m less-than-or-equal-to linear constraints

Feasible region and objective function:

- $F = \{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{A}\,\boldsymbol{x} \leq \boldsymbol{b},\ \boldsymbol{x} \geq 0\}$ is a polyhedron
- $f(\boldsymbol{x}) = \boldsymbol{c}'\boldsymbol{x}$ is a linear function, with gradient $\boldsymbol{c}$

Summary (2/2)

Lower bounds:

- Every feasible solution $\bar{\boldsymbol{x}}(\,LP\,) \in F$ gives $LB(\bar{\boldsymbol{x}}) = \sum_{j=1}^{n} c_j\,\bar{x}_j \leq z(\,LP\,)$

Graphical solution method:

- With two variables: move the level sets, perpendicular to the gradient, in the direction of the gradient, until the last points of F
- First example: $\tilde{\boldsymbol{x}}(\,LP\,) = (5, \frac{7}{2})$, $z(\,LP\,) = 32$; second example: $\tilde{\boldsymbol{x}}(\,LP\,) = (1, 1)$, $z(\,LP\,) = -6$

Standard maximization form

Introduction: Standard maximization form

- We now write an LP problem with **equality** linear constraints instead of less-than-or-equal-to ones: the **standard maximization form**
- The data are the same ($\boldsymbol{c}$, $\boldsymbol{A}$, $\boldsymbol{b}$), and so are the definitions of feasible region, objective function and lower bound
- The example has three variables: we draw it in the space $\mathbb{R}^3$, where the level sets are planes
- The standard form is the form used by the simplex method, later in the course

Definition 4: Linear Programming problems in standard maximization form (1/3)

Given:

1. *an objective-function vector* $\boldsymbol{c} \in \mathbb{R}^n$
2. *a constraint matrix* $\boldsymbol{A} \in \mathbb{R}^{m \times n}$
3. *a right-hand-side (RHS) vector* $\boldsymbol{b} \in \mathbb{R}^m$ *with* m *rows*

$$\underbrace{\begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}}_{\boldsymbol{c}} \in \mathbb{R}^n \quad \underbrace{\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}}_{\boldsymbol{A}} \in \mathbb{R}^{m \times n} \quad \underbrace{\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}}_{\boldsymbol{b}} \in \mathbb{R}^m$$

Definition: Linear Programming problems in standard maximization form (2/3)

*a **Linear Programming problem** LP in **standard maximization form** with n non-negative variables*

$$\underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ \vdots \\ x_n \end{pmatrix}}_{\boldsymbol{x}} \in \mathbb{R}^n_{\geq 0}$$

and m equality linear constraints is the optimization problem of the next slide.

Definition: Linear Programming problems in standard maximization form (3/3)

The LP problem in standard maximization form is the following optimization problem:

$$\max \underbrace{\sum_{j=1}^{n} c_j x_j}_{\boldsymbol{c}'\boldsymbol{x}}$$

$$\textit{subject to} \quad \underbrace{\sum_{j=1}^{n} a_{ij} x_j = b_i, \qquad \forall i \in \{1, 2, \ldots, m\},}_{\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}}$$

$$\underbrace{x_j \geq 0, \qquad \forall j \in \{1, 2, \ldots, n\},}_{\boldsymbol{x}\geq 0}$$

where $0 = (0, 0, \ldots, 0)'$ *is a (column) vector of* n *zeros.*

Canonical and standard forms

- The only difference with the canonical maximization form is in the constraints:

$$\boldsymbol{A}\,\boldsymbol{x} \leq \boldsymbol{b} \quad \text{(canonical)} \qquad\qquad \boldsymbol{A}\,\boldsymbol{x} = \boldsymbol{b} \quad \text{(standard)}$$

- In both forms the variables are non-negative and the objective function $\boldsymbol{c}'\boldsymbol{x}$ is maximized

Let us build an LP problem in standard maximization form from its data $\boldsymbol{c}$, $\boldsymbol{A}$ and $\boldsymbol{b}$.

Example: LP problem in standard maximization form (1/2)

Consider the following input data of an LP problem with $n = 3$ and $m = 1$:

$$\boldsymbol{c} = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} \qquad \boldsymbol{A} = \begin{pmatrix} 2 & 3 & 1 \end{pmatrix} \qquad \boldsymbol{b} = \begin{pmatrix} 4 \end{pmatrix}$$

The numerical entries of the two column vectors and the matrix are

$$c_1 = 2,\ c_2 = 2,\ c_3 = 1,\ a_{11} = 2,\ a_{12} = 3,\ a_{13} = 1,\ b_1 = 4.$$

Example: LP problem in standard maximization form (2/2)

The problem involves $n = 3$ non-negative variables $\boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$.

The associated LP problem in standard maximization form with $n = 3$ non-negative variables and $m = 1$ equality linear constraints reads as follows:

$$
\begin{array}{rrrrrrrrrr}
\max & 2 & x_1 & + & 2 & x_2 & + & x_3 & & \\
\text{subject to} & 2 & x_1 & + & 3 & x_2 & + & x_3 & = & 4 \\
 & & x_1, & & & x_2, & & x_3 & \geq & 0
\end{array}
$$

The feasible region and the objective function are defined as in the canonical maximization form:

only the constraints that define F change.

Definition 5: feasible region and objective function (1/2)

- *The **feasible region** of an LP problem in standard maximization form is the **polyhedron** F defined as follows:*

$$F = \left\{ \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}}_{\boldsymbol{x}} \in \mathbb{R}^n : \underbrace{\sum_{j=1}^{n} a_{ij} x_j = b_i, \quad \forall i \in \{1, 2, \ldots, m\},}_{\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}} \; \underbrace{x_j \geq 0, \quad \forall j \in \{1, 2, \ldots, n\}}_{\boldsymbol{x}\geq 0} \right\}.$$

The feasible region is defined by the equality linear constraints and the non-negativity constraints.

Definition: feasible region and objective function (2/2)

- *The **objective function** of an LP problem in standard maximization form to be maximized is the (scalar) **linear function** f defined as follows:*

$$f : \mathbb{R}^n \to \mathbb{R}, \qquad f : \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}}_{\boldsymbol{x}} \mapsto \underbrace{\sum_{j=1}^{n} c_j x_j}_{\boldsymbol{c'x}}$$

The example has three variables:
we draw it in the space (x_1, x_2, x_3),
where the level sets of the objective function are **planes**.

Example: objective function (1/2)

The objective function of the LP problem in standard maximization form is the scalar linear function:

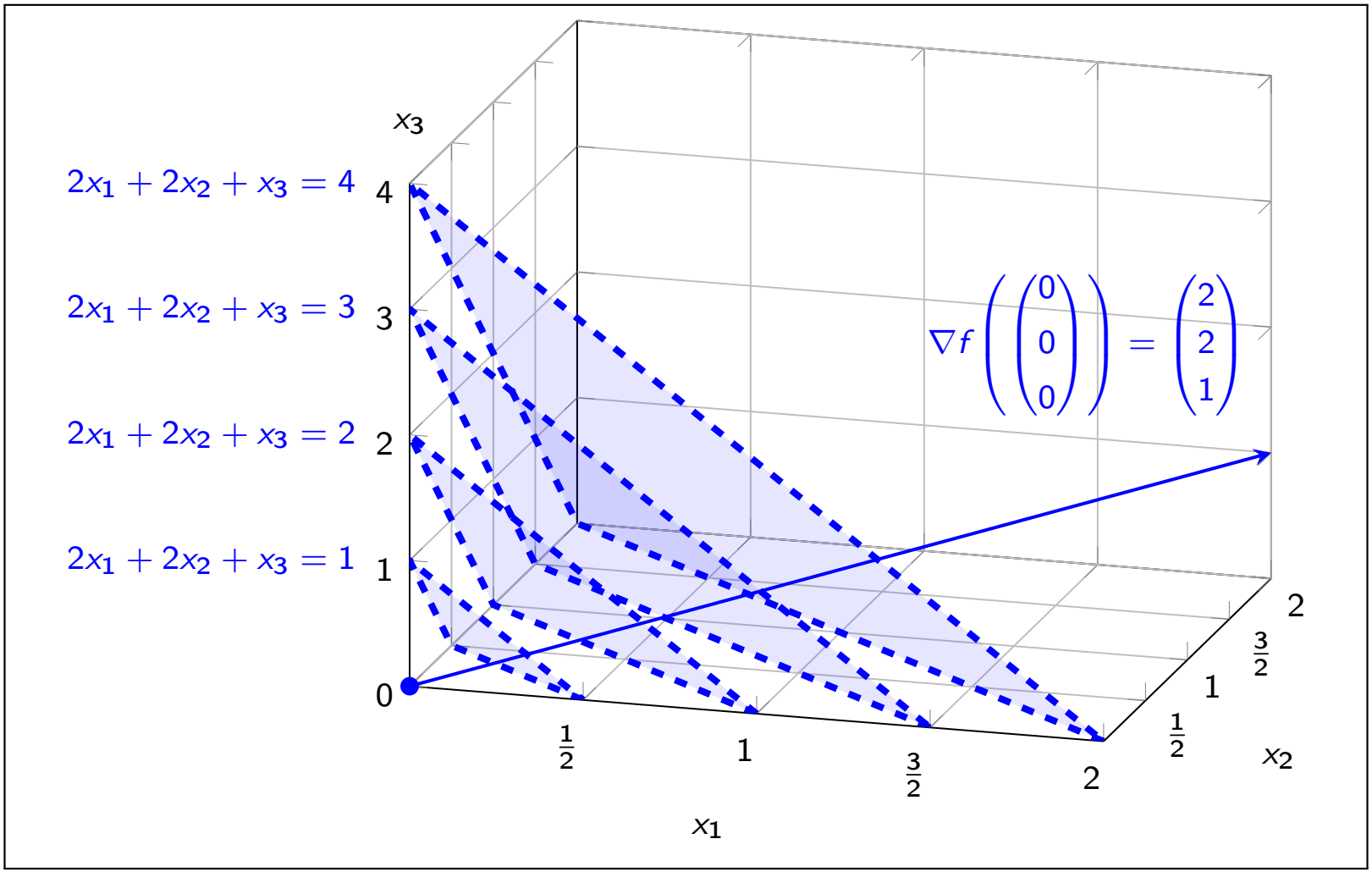
$$f : \mathbb{R}^3 \to \mathbb{R}, \quad f : \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \mapsto 2\,x_1 + 2\,x_2 + x_3$$

and its gradient is

$$\nabla f \left(\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \right) = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}.$$

- In the next slide: the gradient drawn from the origin and the level sets $2x_1 + 2x_2 + x_3 = 1, 2, 3, 4$ (dashed blue), restricted to $x_1, x_2, x_3 \geq 0$

Example: objective function (2/2)



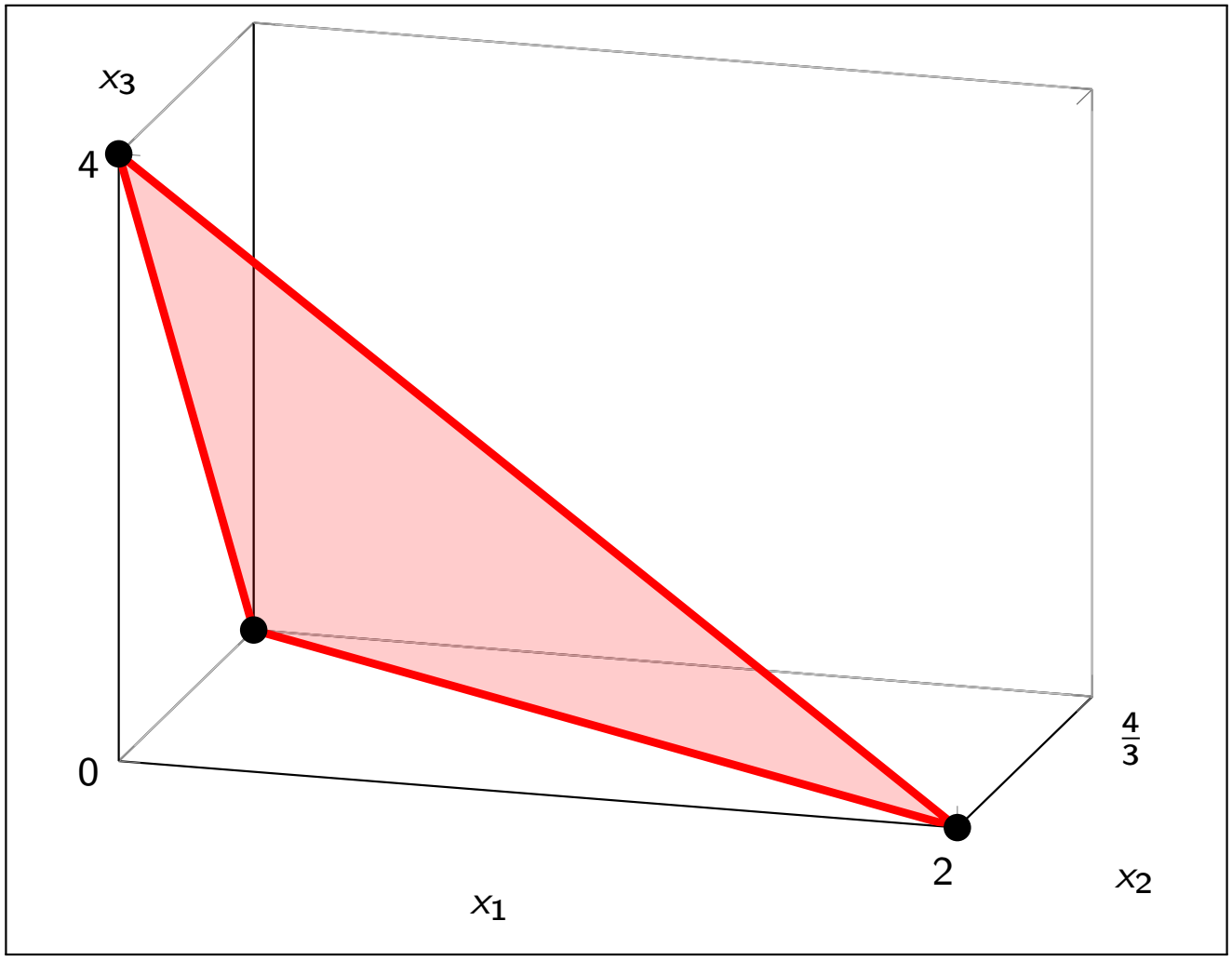
Example: feasible region (1/2)

The feasible region of the LP problem in standard maximization form is the polyhedron:

$$F = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in \mathbb{R}^3 : \begin{aligned} 2x_1 + 3x_2 + x_3 &= 4, \\ x_1 &\geq 0, \\ x_2 &\geq 0, \\ x_3 &\geq 0 \end{aligned} \right\}.$$

- In the next slide: F is the red triangle of the plane $2x_1 + 3x_2 + x_3 = 4$ with vertices (black) $(2, 0, 0)$, $(0, \frac{4}{3}, 0)$ and $(0, 0, 4)$

Example: feasible region (2/2)



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As in the canonical maximization form,
every feasible solution gives a lower bound
on the optimal value $z(\,LP\,)$.

Definition 6: feasible solution and lower bound

- *Given*

$$\underbrace{\begin{pmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \vdots \\ \bar{x}_n \end{pmatrix}}_{\bar{\mathbf{x}}(\,LP\,)} \in F$$

*a **feasible solution** of the LP problem in standard maximization form (a point satisfying the equality constraints and the non-negativity constraints), we can compute a **lower bound** $LB(\bar{\mathbf{x}})$ on the optimal value $z(\,LP\,)$ as follows:*

$$LB(\bar{\mathbf{x}}) = \sum_{j=1}^{n} c_j \, \bar{x}_j$$

Lower bounds and optimal value

For every feasible solution $\bar{\boldsymbol{x}}(\, LP\,) \in F$ of an LP problem in standard maximization form, we have

$$LB(\bar{\boldsymbol{x}}) \leq z(\, LP\,)$$

Example: feasible solutions and lower bounds (1/2)

Consider the LP problem in standard maximization form. A feasible solution is for example

$$\bar{\boldsymbol{x}}(\,LP\,) = \begin{pmatrix} 1 \\ \frac{1}{3} \\ 1 \end{pmatrix} \qquad (\text{where } \bar{x}_1 = 1,\ \bar{x}_2 = \tfrac{1}{3} \text{ and } \bar{x}_3 = 1)$$

since:

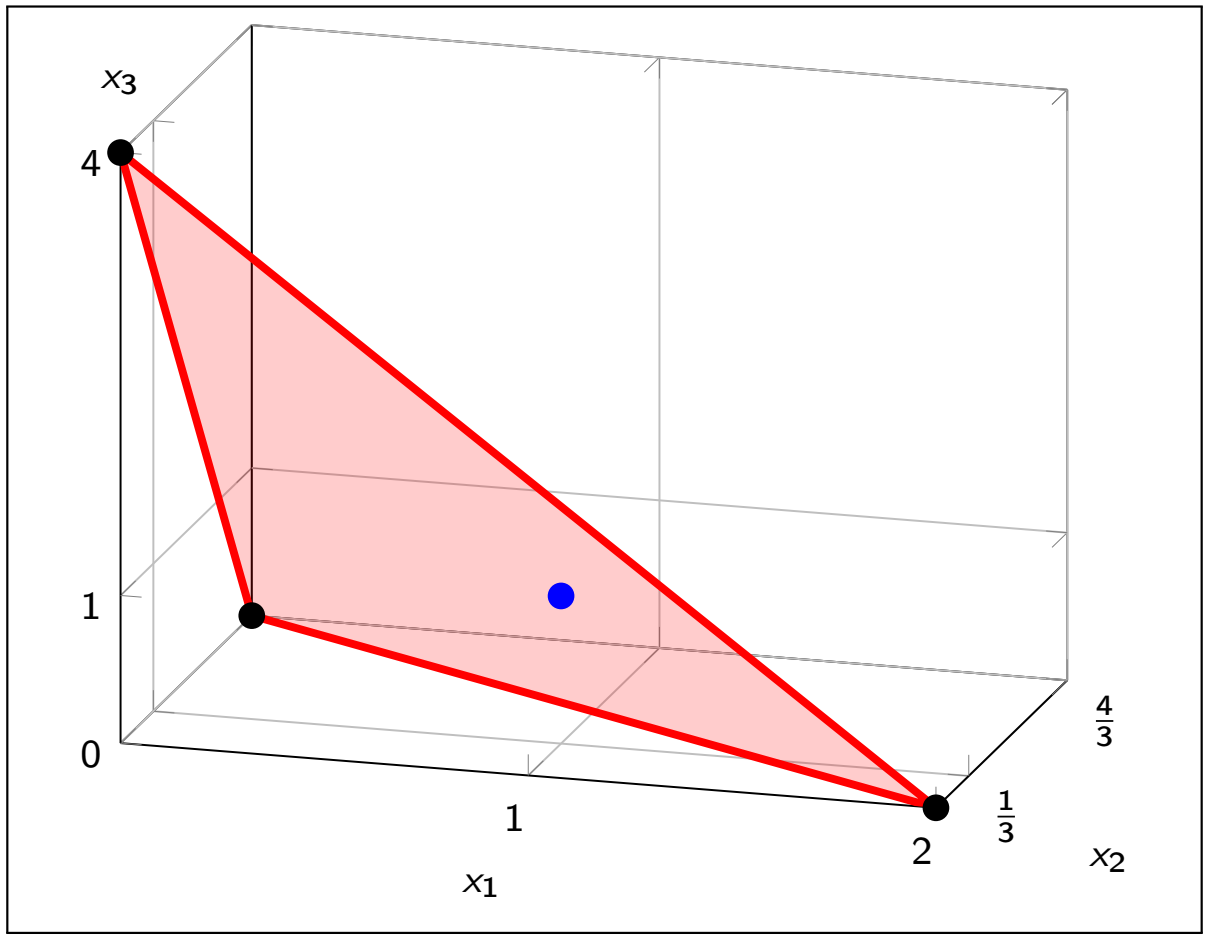
$$2 \cdot 1 + 3 \cdot \tfrac{1}{3} + 1 = 4, \qquad 1 \geq 0, \qquad \tfrac{1}{3} \geq 0, \qquad 1 \geq 0.$$

Accordingly, a lower bound $LB(\bar{\boldsymbol{x}})$ on the optimal value $z(\,LP\,)$ of the LP problem is:

$$LB(\bar{\boldsymbol{x}}) = 2 \cdot 1 + 2 \cdot \tfrac{1}{3} + 1 = \frac{11}{3}.$$

Example: feasible solutions and lower bounds (2/2)

- The feasible solution $(1, \frac{1}{3}, 1)$ (blue) lies in the triangle F



Graphical solution method

An optimal solution of an LP problem in standard maximization form with two or three variables can be **graphically** determined using the direction of the objective function gradient and the level sets perpendicular to the gradient.

- With three variables the level sets are planes: we move them in the direction of the gradient until the last points of F

Example: graphical solution method (1/3)

For the LP problem in standard maximization form:

$$\begin{array}{rrrrrrrrrr} \max & 2 & x_1 & + & 2 & x_2 & + & x_3 & & \\ \text{subject to} & 2 & x_1 & + & 3 & x_2 & + & x_3 & = & 4 \\ & & x_1, & & & x_2, & & x_3 & \geq & 0 \end{array}$$

all optimal solutions lie on the line segment

$$\left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in \mathbb{R}^3 \ : \ x_2 = 0,\ 2x_1 + x_3 = 4,\ x_1 \geq 0,\ x_3 \geq 0 \right\},$$

i.e., on the segment joining the points $\begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix}$.

Example: graphical solution method (2/3)

- Taking for instance the optimal solution

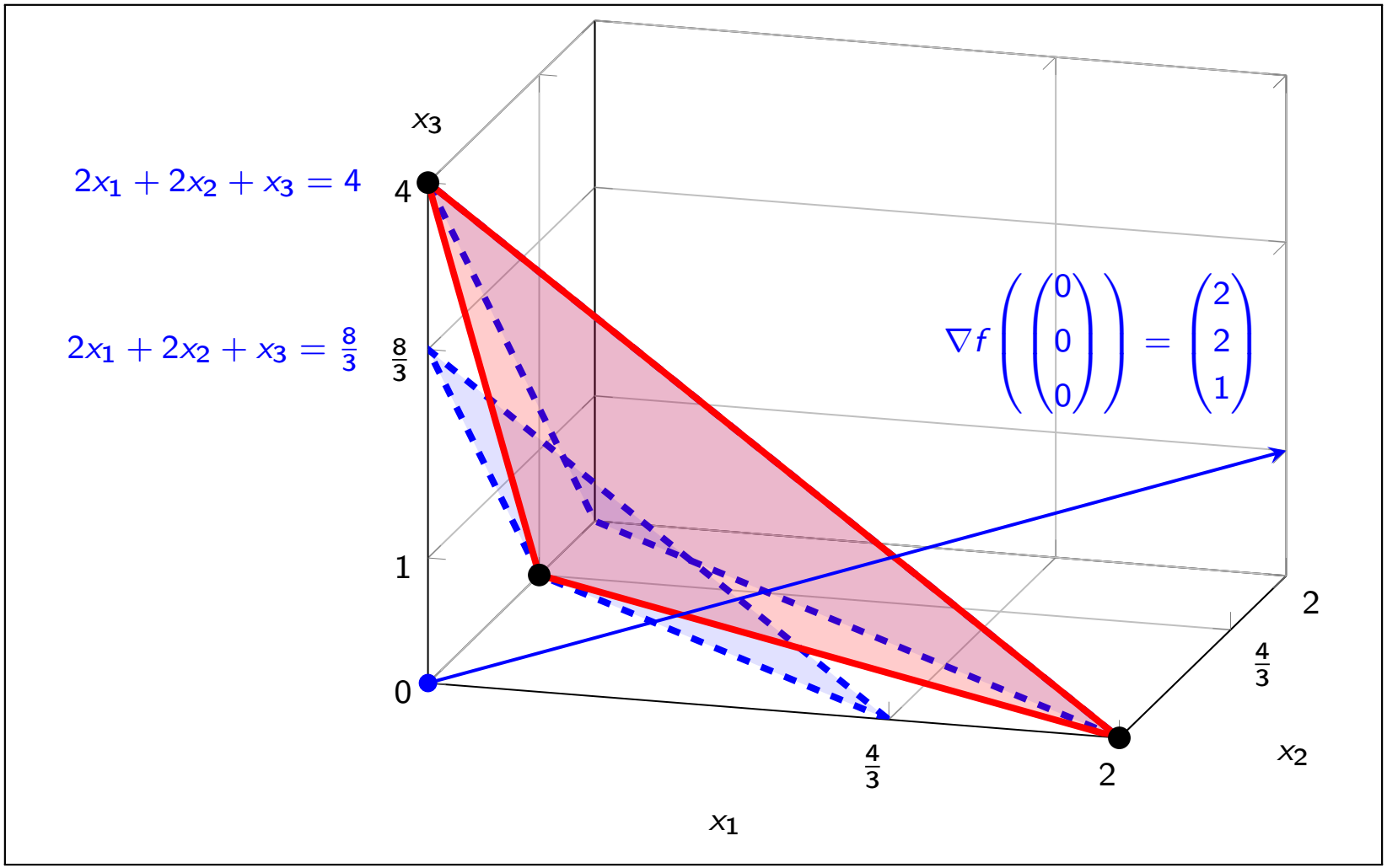
$$\tilde{\boldsymbol{x}}(\,LP\,) = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} \qquad \text{(where } \tilde{x}_1 = 2,\ \tilde{x}_2 = 0 \text{ and } \tilde{x}_3 = 0\text{)},$$

  the optimal value is:

$$z(\,LP\,) = 2\,\tilde{x}_1 + 2\,\tilde{x}_2 + \tilde{x}_3 = 2 \cdot 2 + 2 \cdot 0 + 0 = 4.$$

- In the next slide: the level sets $2x_1 + 2x_2 + x_3 = \frac{8}{3}$ and $2x_1 + 2x_2 + x_3 = 4$ (dashed blue) and the feasible region F (red); the level set of value 4 touches F along the whole segment of the optimal solutions

Example: graphical solution method (3/3)



Summary

Standard maximization form:

- max $\boldsymbol{c}'\boldsymbol{x}$ subject to $\boldsymbol{A}\,\boldsymbol{x} = \boldsymbol{b}$, $\boldsymbol{x} \geq 0$: n non-negative variables and m equality linear constraints
- $F = \{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{A}\,\boldsymbol{x} = \boldsymbol{b},\ \boldsymbol{x} \geq 0\}$ is a polyhedron

Lower bounds and graphical solution method:

- Every feasible solution gives $LB(\bar{\boldsymbol{x}}) \leq z(\,LP\,)$
- With two or three variables the problem can be solved graphically
- An LP problem can have infinitely many optimal solutions: in the example, a whole segment, with $z(\,LP\,) = 4$

Summary of the chapter (1/2)

Two forms of LP problems:

- Both maximize $\boldsymbol{c}'\boldsymbol{x}$ over n non-negative variables, with data $\boldsymbol{c}$, $\boldsymbol{A}$, $\boldsymbol{b}$
- Canonical maximization form: m constraints $\boldsymbol{A}\,\boldsymbol{x} \leq \boldsymbol{b}$
- Standard maximization form: m constraints $\boldsymbol{A}\,\boldsymbol{x} = \boldsymbol{b}$

Feasible region and objective function:

- The feasible region F is a polyhedron; the objective function $f(\boldsymbol{x}) = \boldsymbol{c}'\boldsymbol{x}$ is linear, with gradient $\boldsymbol{c}$

Summary of the chapter (2/2)

Lower bounds:

- Every feasible solution $\bar{\boldsymbol{x}}(\, LP\,) \in F$ gives a lower bound $LB(\bar{\boldsymbol{x}}) = \boldsymbol{c}'\bar{\boldsymbol{x}} \leq z(\, LP\,)$

Graphical solution method:

- With two (or three) variables, the level sets perpendicular to the gradient are moved in the direction of the gradient until the last points of F
- The optimal solution can be a single vertex or a whole segment of optimal solutions